Practice workbook · Finding the equation of a normal

Finding the gradient of the curve at a point, turning it into the normal gradient with the negative reciprocal and writing the normal's equation.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-05

1. Question 9 · part whole · 5 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part whole only.

9 Find an equation of the normal to the curve with equation

$$y = (x^3 - 2x)e^{(1-x)}$$

at the point on the curve with coordinates $(1, -1)$

(5)

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1. Question 9 · part whole · 5 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part whole only.

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Question 9 continued

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(Total for Question 9 is 5 marks)

1. Question 9 · part whole · 5 marks · Working and answer

Working and answer

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2. Question 4 · part whole · 10 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

4 $f(x) = e^{3x}\sqrt{1+2x}$

(a) Show that $f'(x) = \dfrac{2e^{3x}(2+3x)}{\sqrt{1+2x}}$

(4)

(b) Find an equation of the normal to the curve with equation $y = f(x)$ at the point on the curve where $x = 0$

Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(6)

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2. Question 4 · part whole · 10 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 10 marks)

2. Question 4 · part whole · 10 marks · Working and answer

Working and answer

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3. Question 8 · part c · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

8 Given that $2xy + 5y = e^x$

(a) show that $\frac{dy}{dx} = \frac{y(2x+3)}{2x+5}$

(5)

(b) find the value of $\frac{dy}{dx}$ when $x = 0$

(2)

(c) find an equation of the normal to the curve with equation $2xy + 5y = e^x$ at the point where $x = 0$
Give your answer in the form $px + qy + r = 0$ where p, q and r are integers.

(3)

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3. Question 8 · part c · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

Question 8 continued

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3. Question 8 · part c · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

Question 8 continued

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3. Question 8 · part c · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 10 marks)

3. Question 8 · part c · 3 marks · Working and answer

Working and answer

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4. Question 5 · part b · 5 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

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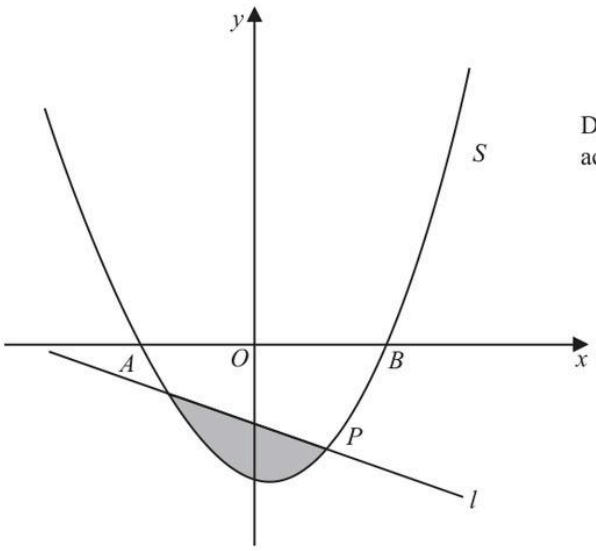


Figure 1

Figure 1 shows part of the curve S with equation $y = px^2 + qx + r$ where p, q and r are constants.

The points A, B and P with coordinates $(-2, 0)$, $(6, 0)$ and $(4, -6)$ respectively lie on S

(a) Show that an equation of S is $y = \frac{x^2}{2} - 2x - 6$

(3)

The line l is the normal to S at the point P

(b) Show that an equation of l is $2y + x + 8 = 0$

(5)

The finite region shown shaded in Figure 1 is bounded by S and l

(c) Use algebraic integration to find the exact area of the shaded region.

(7)

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P 7 3 5 8 6 A 0 1 2 3 2

4. Question 5 · part b · 5 marks · Working and answer

Working and answer

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5. Question 6 · part c · 3 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part c only.

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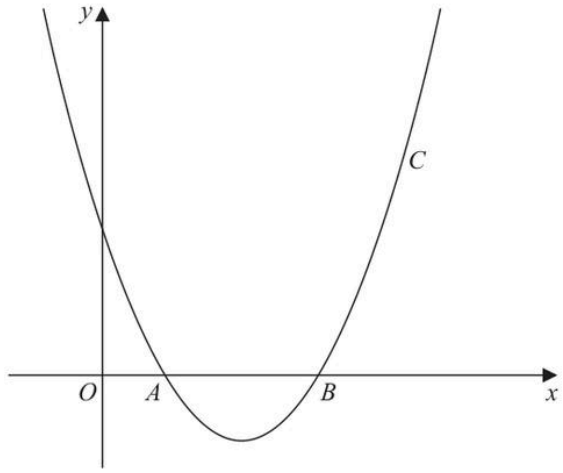


Diagram **NOT** accurately drawn

Figure 3

The curve C with equation $y = x^2 - 5x + 4$ crosses the x-axis at the points A and B, as shown in Figure 3

(a) Find the coordinates of A and the coordinates of B.

(3)

The tangent to C at A meets the tangent to C at B at the point T.

(b) Find the coordinates of T.

(6)

The normal to C at A meets the normal to C at B at the point N.

(c) Find the coordinates of N.

(3)

(d) Find the area of the quadrilateral $ATBN$.

(3)

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5. Question 6 · part c · 3 marks · Working and answer

Working and answer

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6. Question 6 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

6 The curve C has equation

$$y = 4\sqrt{x}$$

The point A on C has coordinates (9, 12)

The tangent to C at the point A meets the x-axis at the point T

(a) Find the coordinates of T

(5)

The normal to C at the point A meets the x-axis at the point N

(b) Find the coordinates of N

(4)

(c) Calculate the area of triangle ATN

(2)

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6. Question 6 · part b · 4 marks · Working and answer

Working and answer

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7. Question 11 · part b · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

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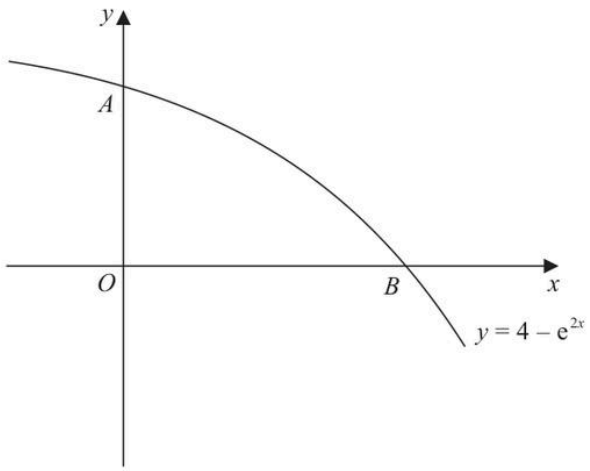


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = 4 - e^{2x}$
The curve C crosses the y-axis at the point A and the x-axis at the point B.

(a) (i) Write down the y coordinate of point A.

(ii) Show that the x coordinate of B is $x = \ln 2$

(3)

The line l is the normal to C at the point B.

(b) Find an equation for l, giving your answer in the form $y = mx + c$

(4)

The finite region R is bounded by C, l and the y-axis.

(c) Using calculus, find the area of R.
Give your answer to one decimal place.

(7)

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7. Question 11 · part b · 4 marks · Working and answer

Working and answer

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8. Question 7 · part a · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part a only.

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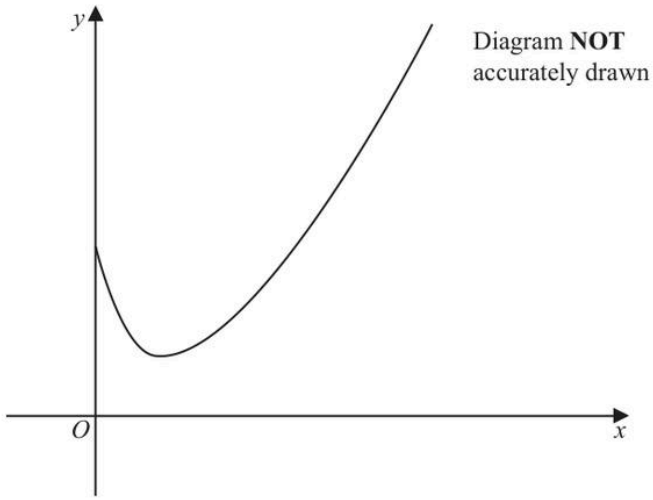


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Figure 1

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{x^2}{4} - 3\sqrt{x} + 8$$

The point P lies on C and has coordinates $(4, a)$

(a) Show that $a = 6$

(1)

The line L is the normal to C at the point P

(b) Show that an equation of L is $5y + 4x - 46 = 0$

(6)

The finite region R is bounded by the curve C, the line L, the x-axis and the line with equation $x = 1$

(c) Use calculus to find the exact area of R

(6)

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8. Question 7 · part a · 1 mark · Working and answer

Working and answer

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9. Question 7 · part b · 6 marks

2024 JUN · Paper 01

Further practice · Attempt part b only.

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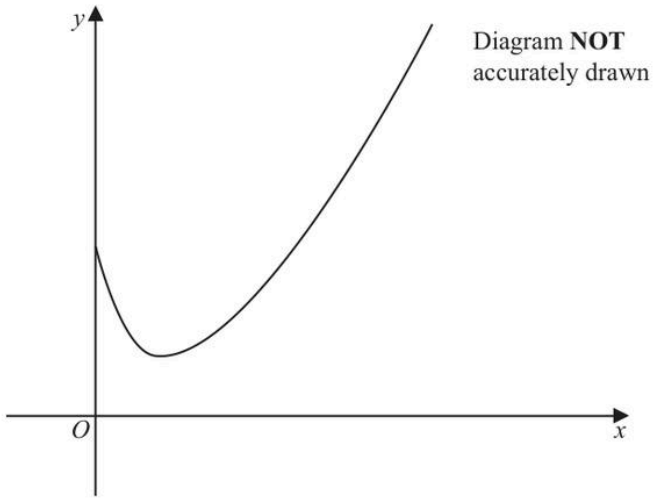


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The point P lies on C and has coordinates $(4, a)$

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(1)

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(b) Show that an equation of L is $5y + 4x - 46 = 0$

(6)

The finite region R is bounded by the curve C, the line L, the x-axis and the line with equation $x = 1$

(c) Use calculus to find the exact area of R

(6)

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9. Question 7 · part b · 6 marks · Working and answer

Working and answer

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10. Question 7 · part b · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

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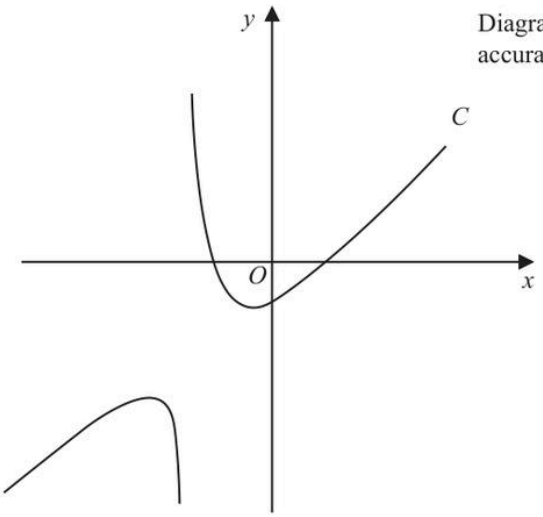


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Figure 2

Figure 2 shows a sketch of part of the curve C with equation

$$y = \frac{x^2 - 1}{4x + 5} \quad \text{where} \quad x \neq -\frac{5}{4}$$

(a) Write down the equation of the asymptote to C that is parallel to the y-axis.

(1)

The line l is the normal to C at the point where $x = -1$

(b) Find an equation of l

(7)

The line l meets C again at the point D

(c) Find the coordinates of D

(6)

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14

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10. Question 7 · part b · 7 marks · Working and answer

Working and answer

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11. Question 4 · part c · 6 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

4 The curve C has equation $y = 2x^2 + px + q$ where p and q are integers.

The curve C has a stationary point at $(3, -5)$

(a) Show that $p = -12$ and find the value of q

(4)

(b) State, giving a reason, the nature of the stationary point.

(1)

(c) Find an equation of the normal to C at the point on C where $x = 1$

Give your answer in the form $ax + by + c = 0$

(6)

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11. Question 4 · part c · 6 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 11 marks)

11. Question 4 · part c · 6 marks · Working and answer

Working and answer

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12. Question 10 · part d · 6 marks

2023 JUN · Paper 02

Further practice · Attempt part d only.

10 The curve C with equation $y = \dfrac{6 - 3x}{x - 4}$ where $x \neq 4$, crosses the x-axis at the point P and the y-axis at the point Q

(a) Find the coordinates of

(i) P (ii) Q

(2)

(b) Write down an equation of the asymptote to C which is

(i) parallel to the y-axis (ii) parallel to the x-axis

(2)

(c) Sketch C showing clearly the asymptotes and the coordinates of the points P and Q

(3)

The line L is the normal to C at the point on C where $x = 2$

(d) Find an equation of L

(6)

The line L intersects C again at the point R

(e) Find the x coordinate of R

(3)

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12. Question 10 · part d · 6 marks · Working and answer

Working and answer

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13. Question 8 · part whole · 8 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

8 The curve C has equation $y = 2x^2 - \sin x$

The point A on C has x coordinate π

Show that an equation of the normal to C at the point A is

$$x + (4\pi + 1)y - \pi(8\pi^2 + 2\pi + 1) = 0$$

(8)

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13. Question 8 · part whole · 8 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 8 continued

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(Total for Question 8 is 8 marks)

13. Question 8 · part whole · 8 marks · Working and answer

Working and answer

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14. Question 9 · part b · 7 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

9 Given that $f(x) = (1+x^2)^4$

(a) show that $f'(x) = 8x(1+x^2)^3$ **(2)**

The curve C has equation $y = \dfrac{\sin 3x}{(1+x^2)^4}$

The point A on C has x coordinate $\dfrac{2\pi}{3}$

(b) Show that the gradient of the normal to C at A is

$$-\frac{1}{3}\left(1+\left(\frac{2\pi}{3}\right)^2\right)^4$$

(7)

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14. Question 9 · part b · 7 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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Question 9 continued

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14. Question 9 · part b · 7 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 9 continued

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14. Question 9 · part b · 7 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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Question 9 continued

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(Total for Question 9 is 9 marks)

14. Question 9 · part b · 7 marks · Working and answer

Working and answer

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